Premium Integration. Not out the box and needs to be customised within a secure Next Generation Network. Customisation is storing certain sign offs in blockchain and sharing them in the network.

Integrating the supply chain software modules with a Hyperledger Fabric blockchain network can provide enhanced transparency, traceability, and security across the supply chain. Here's how the integration can be achieved:

1. **Inventory Management Module Integration:**

- The inventory management module can store critical supply chain data, such as product information, quantity, location, and timestamps, on the Hyperledger Fabric blockchain.

- This allows for immutable record-keeping and real-time visibility of inventory movements across the supply chain.

- Blockchain-based smart contracts can be used to automate inventory replenishment and trigger notifications based on predefined rules.

The warehouses shall make use of the Inventory module within the Blockchain network, while organisation maintain their own Inventory within the core Machuzu ERP database.

2. **Order Management Module Integration:**

- The order management module can leverage the Hyperledger Fabric blockchain to record and track order transactions, from order placement to fulfillment.

- Smart contracts can be used to automate order processing, shipment tracking, and invoice generation, ensuring transparency and reducing manual intervention.

- Blockchain-based proofs of delivery can be used to validate and confirm order fulfillment, enhancing trust between parties.

Machuzu ERP contains all the Order Management modules required. The software currently can manage the process from Requisition to Delivery of goods.

3. **Procurement and Supplier Management Module Integration:**

- The procurement and supplier management module can use the Hyperledger Fabric blockchain to maintain a secure, tamper-proof register of supplier contracts, purchase orders, and invoices.

- Smart contracts can be used to manage the supplier onboarding process, automate contract negotiations, and enforce compliance with agreed-upon terms.

- Supplier performance data can be stored on the blockchain, enabling transparent and objective supplier evaluation and selection.

Machuzu ERP shall continue to process the Purchase Orders and Invoices, then post them to the Blockchain network once finalised; for them to be accessible within the Supply Chain processes.

4. **Integration Architecture:**

- The supply chain software modules can be integrated with the Hyperledger Fabric blockchain network through the use of API gateways or event listeners.

- These integration points allow the software modules to interact with the blockchain, triggering transactions, querying data, and executing smart contracts as needed.

- The integration can be designed to be bi-directional, where the software modules can both read from and write to the blockchain, ensuring a seamless flow of information.

5. **Data Synchronization and Reconciliation:**

- To maintain data consistency and integrity, the supply chain software modules and the Hyperledger Fabric blockchain network must be synchronized.

- Automated data reconciliation processes can be implemented to ensure that any updates or changes made in one system are reflected in the other, preventing discrepancies.

By integrating the supply chain software modules with a Hyperledger Fabric blockchain network, organizations can achieve enhanced transparency, traceability, and trust across their supply chain operations. This integration can help reduce risks, improve efficiency, and enable better collaboration among supply chain partners.